Differential Leveling

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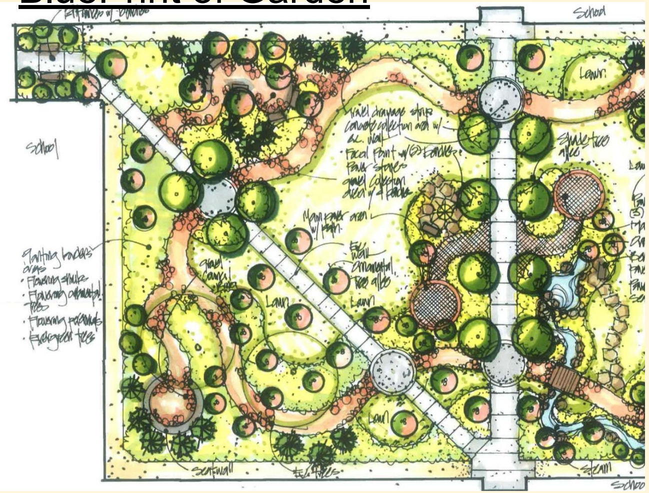
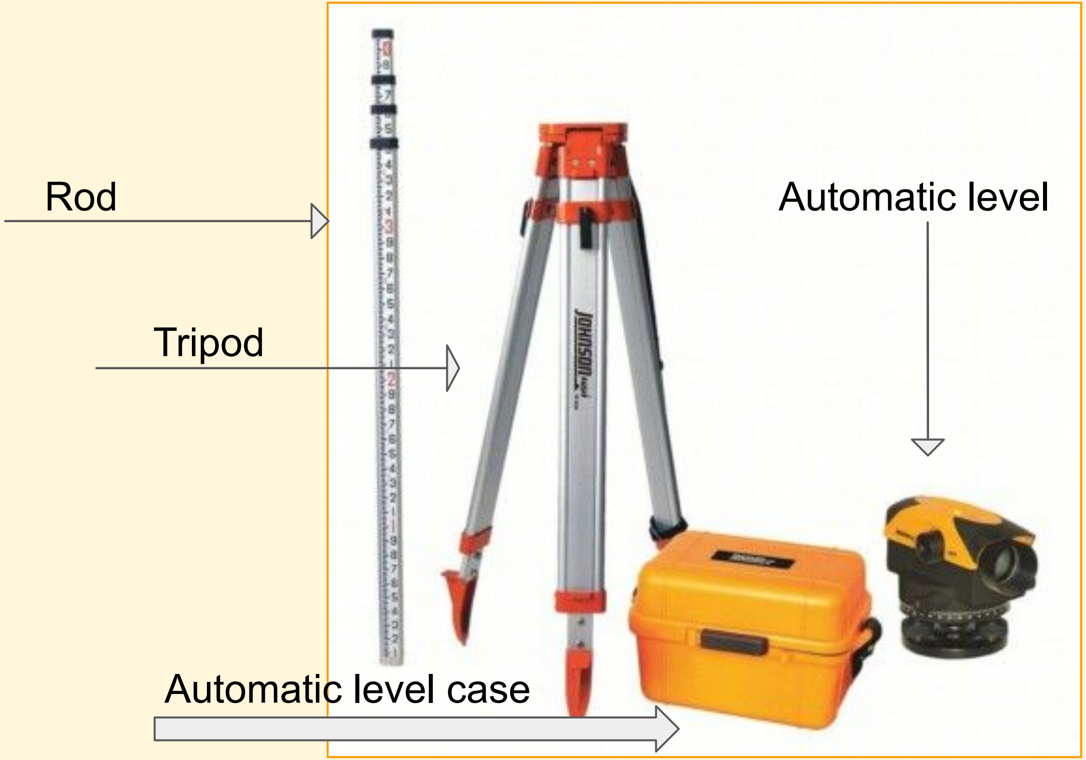
Description

Our challenge was to map out as much of the garden using an automatic level, rod, and tripod to estimate the distance of the landmarks in the garden from a certain area we got to pick out.

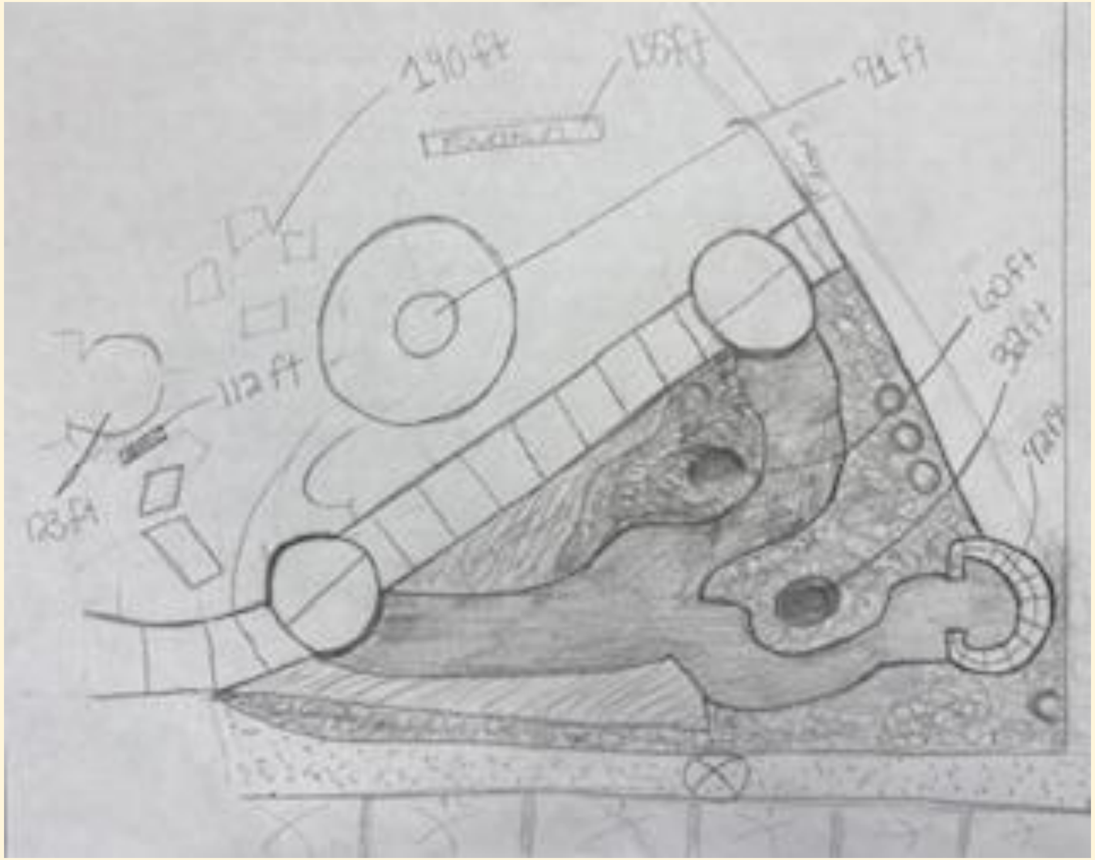
<u>BluePrint of Garden</u>





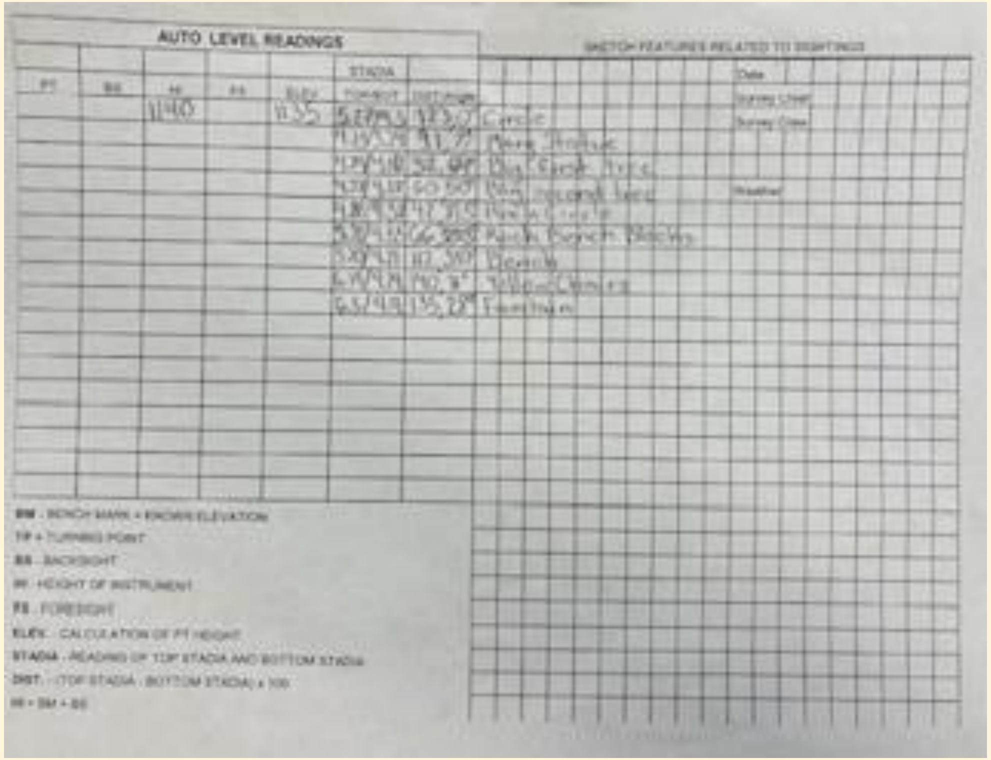
Pictures

Sketches



CAD

Data



AUTO LEVEL READINGS

SKETCH FEATURES RELATED TO SIGHTINGS

PT	BS	HI	FS	ELEV	STADIA	DISTANCE	
		1140		1135	[illegible]	[illegible]	Circle

BM - BENCH MARK + KNOWN ELEVATION

TP = TURNING POINT

BS - BACKSIGHT

HI - HEIGHT OF INSTRUMENT

FS - FORESIGHT

ELEV. - CALCULATION OF PT HEIGHT

STADIA - READING OF TOP STADIA AND BOTTOM STADIA

DIST. - (TOP STADIA - BOTTOM STADIA) x 100

HI = BM + BS

Calculations

Tree 1: 32ft

Tree 2: 60ft

Circle Bench: 42ft

Mary Statue: 91ft

Fountain: 136ft

Rock Bench: 112ft

Path Circle: 123ft

Yellow Chairs: 140ft

Conclusion

R/W/L

Right

We successfully mapped out the garden and made a nice sketch of it.

Wrong

We were unable to calculate the distance of a drain from where we were measuring from.

Learned

We learned how to map out a garden using an automatic level, rod, and tripod.